you're building dangerous hallucination machines. In healthcare, it could mean recommending harmful treatments. In legal documents, it could completely invert contractual obligations. In content moderation, you could miss the difference between 'violence is acceptable' and 'violence is never acceptable'.

Formatting inconsistency

Extra spaces, tabs, weird formatting — the models don't care. Similarity stays above 0.995. But remove all spaces? Similarity suddenly drops to 0.82.

I hit this issue working with scraped content that had irregular spacing due to poor HTML. We built this beautiful search system for a digital library with thousands of scraped documents. Half the queries returned nothing useful because of inconsistent spacing. The librarians were ready to trash the whole project.

This becomes a real problem with user-generated content, OCR'd documents, or languages that don't use spaces like English does (Thai or Chinese). It also means these models struggle with hashtags, URLs, and product codes — things real people search for every day.

Directional confusion

Embedding models see "The car is to the left of the tree" and "The car is to the right of the tree" as nearly identical — an insanely high 0.98 similarity score.

Would warehouse managers be happy when robots deliver packages to the wrong stations? I'd be pulling my hair out.

Think about cases where perspective and reference frames are essential — navigation directions, spatial relationships, positions in medical procedures, or legal descriptions of accident scenes. These distinctions change the entire meaning based on perspective. We're working with models that can't tell if you're describing something from the front or from behind.

Causal relationship reversal

This one made me laugh, then cry. Embedding models see "If demand increases, prices will rise" and "If demand increases, prices will fall" as practically identical — shocking 0.95 similarity score.

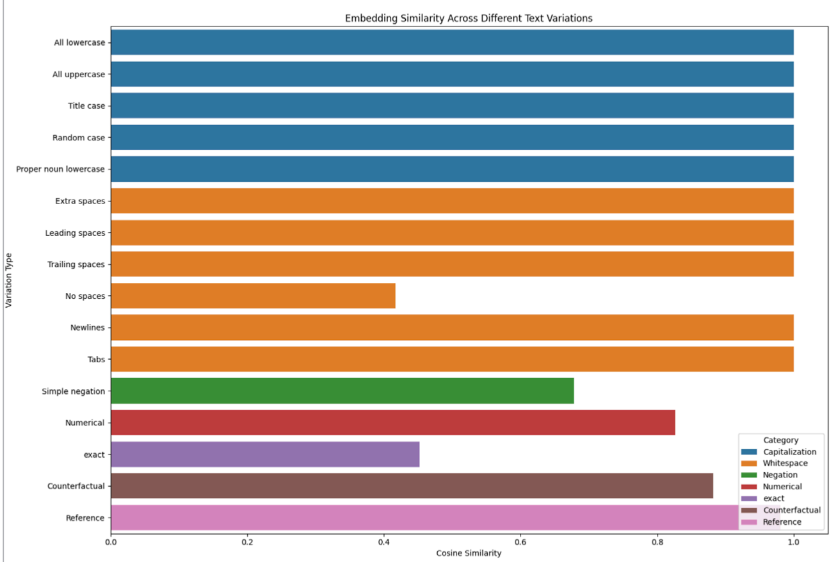
I hit this problem building an analysis system for economic research papers. The algorithm couldn't distinguish between opposing causal relationships. Economists searching for papers about price increases during demand surges got results about price drops during recessions. Would financial analysts making million-dollar investment decisions appreciate getting exactly backwards information? I wouldn't want my retirement managed that way.

Think about all the cases where counterfactual reasoning is critical: economic projections, medical cause-and effect, legal hypotheticals, and engineering failure analysis. When you can't tell 'if X, then Y' from 'if not-X, then not-Y', you fundamentally misunderstand the causal relationship. We're dealing with models that can't grasp basic conditional logic.

Range vs exact value confusion

This one left me speechless. Embedding models see "The product costs between $50-$100" and "The product costs exactly $101" as NEARLY the same thing — mind-boggling 0.98 similarity score.

I discovered this building a price comparison system for an e-commerce client. The search couldn't distinguish between price ranges and exact prices, even when the exact price was outside the specified range. Shoppers with a strict $100 budget searching for products 'under $100' kept seeing items




```
Categories from most to least impact on embeddings:
exact: 0.4520
Negation: 0.6779
Numerical: 0.8264
Counterfactual: 0.8823
Whitespace: 0.9028
Reference: 0.9804
Capitalization: 1.0000
```

Figure 2: all-mpnet-base-v2 embedding score across different test cases